

Dot-Com Calibration, Kappa Severity, and Scenario Mechanics

Theory Slides for the AI Unwind Stress Framework

Prepared for Senior Management

July 2026

Slide 1 - Why dot-com calibration enters the theory

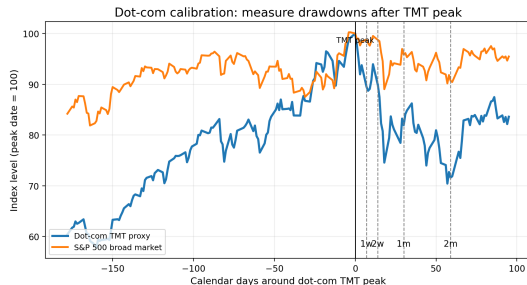
The model needs historical shock magnitudes.

The dot-com episode is used because it is the closest observable analogue of a technology-led equity bubble unwinding through both broad-market stress and sector-specific excess stress.

We define two historical paths:

- P_t^W : broad-market price index at date t ; proxied by the S&P 500.
- P_t^E : technology epicenter price index at date t ; proxied by a dot-com TMT basket.

Purpose: use dot-com to calibrate the size of the shock, not to claim the current AI cycle must repeat dot-com exactly.



Slide 2 - Horizon anchors: how the raw dot-com shocks are measured

For each horizon h , we measure the loss from the dot-com TMT peak t_0 to the date $t_0 + h$.

Historical shock decomposition

$$S_h^W = \log \left(\frac{P_{t_0+h}^W}{P_{t_0}^W} \right)$$

$$S_h^E = \log \left(\frac{P_{t_0+h}^E}{P_{t_0}^E} \right)$$

$$S_h^X = S_h^E - S_h^W$$

Variable labels

- h : risk horizon, such as 1w, 2w, 1m, or 2m.
- S_h^W : broad-market shock over horizon h .
- S_h^E : technology epicenter total shock over horizon h .
- S_h^X : technology excess shock over the market.

Why subtract S_h^W ?

- The TMT fall includes both market-wide and technology-specific losses.
- Subtracting the market leg isolates the excess shock to be applied only to AI-sensitive assets.
- This avoids double counting market risk.

Slide 3 - Kappa calibration: turning one history into scenario severity

Dot-com provides the raw historical excess shock S_h^X . Scenario severity controls how much of that excess shock is applied.

Scenario shock construction

$$M_{h,s} = \alpha_s S_h^W$$

$$A_{h,s} = \kappa_s S_h^X$$

$$T_{h,s} = M_{h,s} + A_{h,s}$$

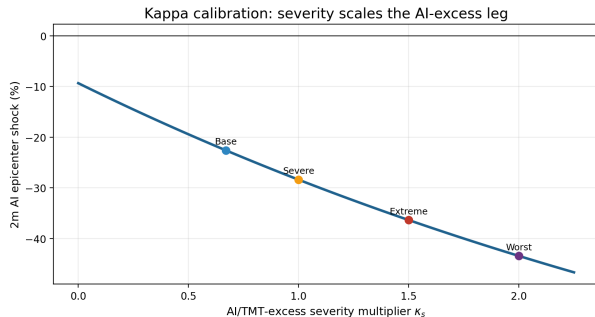
Variable labels

- s : scenario, e.g. mild, base, severe, extreme, worst.
- $M_{h,s}$: market leg used in scenario s and horizon h .
- $A_{h,s}$: AI/TMT-excess leg used in scenario s and horizon h .
- $T_{h,s}$: total technology epicenter shock.

Severity controls

- α_s : market-leg scaling.
- κ_s : AI/TMT-excess severity multiplier.
- In the revised ladder: severe = 1.0, extreme = 1.5, worst = 2.0 for κ_s .

Slide 4 - Kappa mechanics shown visually



How to read the chart

- The horizontal axis is κ_s , the AI/TMT-excess severity multiplier.
- The vertical axis is the resulting 2-month technology epicenter shock.
- A larger κ_s does not mechanically scale everything; it specifically scales the AI/TMT-excess leg.

Scenario	α_s	κ_s	$T_{2m,s}$
Mild	0.50	0.335	-12.0%
Base	1.00	0.671	-22.6%
Severe	1.00	1.0	-28.4%
Extreme	1.00	1.5	-36.4%
Worst	1.00	2.0	-43.5%

Slide 5 - Scenario mechanics: from historical anchors to portfolio shocks

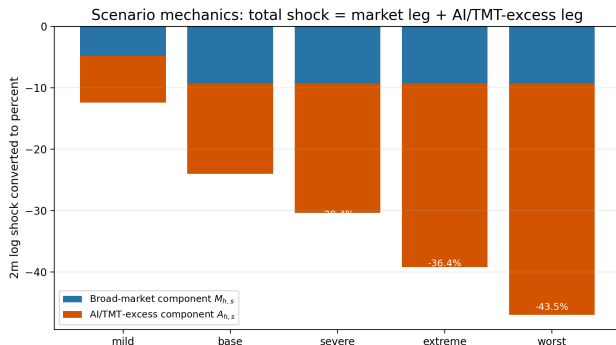
Once $M_{h,s}$ and $A_{h,s}$ are calibrated, each country-industry bucket j receives a stress using its estimated exposures.

Country-industry stress formula

$$R_{j,h,s} = \beta_j^W M_{h,s} + I_j^{AI} A_{h,s}$$

Variable labels

- j : country-industry bucket.
- $R_{j,h,s}$: log stress return for bucket j .
- β_j^W : bucket exposure to the broad market.
- I_j^{AI} : bucket AI-excess intensity.



Slide 6 - Market-cap overlay: scenario stress becomes position-specific

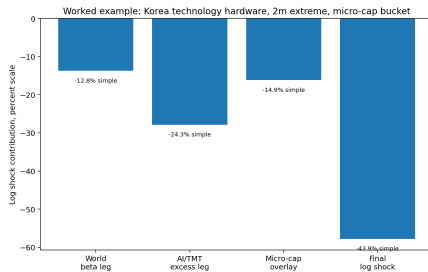
The country-industry shock is then adjusted for market-cap fragility. Smaller-cap names can gap more, face weaker liquidity, and be more affected by forced selling.

Country-industry-market-cap stress formula

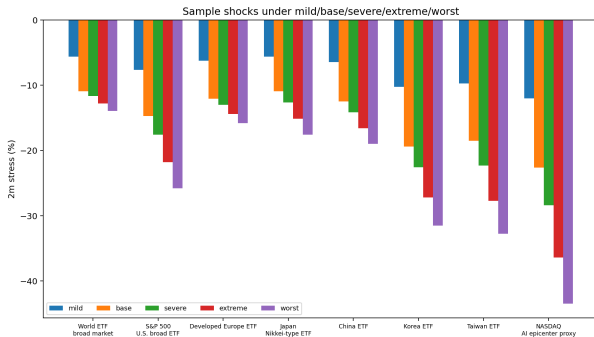
$$R_{j,m,h,s} = R_{j,h,s} + P_{m,h,s}$$

Variable labels

- m : market-cap bucket, from micro to large/mega.
- $P_{m,h,s}$: market-cap penalty overlay for bucket m .
- Large / mega cap is the reference bucket, so $P_{m,h,s} = 0$ there.
- For smaller buckets, $P_{m,h,s} \leq 0$.



Slide 7 - What the scenario system produces



Interpretation

- Broad market proxies receive the market leg plus limited AI-excess intensity.
- NASDAQ / AI epicenter proxies receive much larger AI-excess stress.
- International AI-exposed regions, such as Korea and Taiwan, sit between broad-market and epicenter outcomes.

Why this is coherent

- The results are not manually assigned.
- They are generated by the same formula: exposure times scenario shock.

Slide 8 - Management takeaway

The scenario mechanics are deliberately modular.

- 1 Dot-com history gives raw horizon anchors: S_h^W , S_h^E , and S_h^X .
- 2 Kappa converts those anchors into mild / base / severe / extreme / worst scenarios.
- 3 Bucket exposures convert scenario anchors into country-industry shocks.
- 4 Market-cap penalties convert country-industry shocks into more position-specific stresses.

The complete stress equation is:

$$R_{j,m,h,s} = \beta_j^W M_{h,s} + I_j^{AI} A_{h,s} + P_{m,h,s}.$$

Symbol	Meaning
j	country-industry bucket
m	market-cap bucket
h	horizon: 1w, 2w, 1m, or 2m
s	scenario: mild, base, severe, extreme, or worst
β_j^W	broad-market exposure of bucket j
I_j^{AI}	AI-excess intensity of bucket j
$M_{h,s}$	scenario broad-market shock
$A_{h,s}$	scenario AI/TMT-excess shock
$P_{m,h,s}$	market-cap overlay penalty